



Annual Maintenance Completion Report

(2025-2026)

Generating Unit-4



Mechanical Maintenance Unit

Tala Hydropower plant

Table of Contents

Introduction	3
2. Overview	3
3. Turbine System	4
<i>3.1 Nozzle Injector.....</i>	<i>4</i>
<i>3.2 Runner.....</i>	<i>4</i>
4. Governor and MIV system.....	8
4.1 Governing System	8
<i>4.3 MIV System</i>	<i>9</i>
5. Cooling Water System.....	14
<i>5.1 Clean Cooling Water System.....</i>	<i>14</i>
<i>5.2 Raw Water-Cooling System</i>	<i>14</i>
<i>5.3 Cyclone Separator</i>	<i>15</i>
<i>5.4 Main Heat Exchanger (MHE)</i>	<i>15</i>
Annexure	17

Introduction

The scheduled Check and Repair maintenance of all (6x170MW) generating units and its auxiliaries are carried out annually to ensure smooth and reliable operation of the units during peak generating season. The objective of this maintenance is to keep the generating plants, equipment and other service facilities in an efficient operating condition so that the intended functions are performed satisfactorily and its reliability enhanced. Further, continuous inspection and preventive maintenance of this kind shall help avoid major breakdown. Commonly the Annual maintenance commence from October to May for over seven months period, as the river inflow during this period decreases.

During this period opportunities are taken to attend to all major and minor maintenance works such as replacement of spares, overhauling, servicing, testing and all schedule maintenance required as per O&M on the generating units, its auxiliaries and other installation in the Power House, Butterfly valve (BFV), and steel lined Pressure Shaft during the maintenance period.

This year the annual maintenance began from 17.11.2025 with unit 2. Consecutively annual Maintenance for unit-4 2025-2026 began after the generating unit was taken shut down under PTW No: 7370 on 28.01.2026. Annual maintenance (unit 4) PTW sheet and Schedules are attached as annexure for detailed information. The schedule and exact date of start and completion of annual maintenance of the unit is as given in the table below:

TABLE 1: AM SCHEDULE FOR GENERATING UNIT IV

SN	Unit #	Approved Schedule			Actual Completion			Remarks
		Start	End	No. of Days	Start	End	No of Days	
1.	Unit IV	11.03.2026	25.04.2026	40	28.01.2026	24.02.2026	24	Unit AM works were completed during PS-2 draining for Unit-6 MIV seal replacement, except runner patch weld repair due to time constraints.
2					06.05.2026	14.05.2026	8	Only runner 6020 patch weld repair works were carried out during this period.

The unit was brought to mechanical spin at 09:30 hours on 15.05.2026, and a dry run was initiated on the same day. The machine was officially synchronized at 11:46 hours on 16.05.2026 and subsequently handed over to the operations section after successfully completing the annual maintenance of the unit.

The maintenance activities carried out by Mechanical Maintenance Unit in accordance to annual maintenance plan for the unit is detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

2. Overview

The work plan includes welding repair of Runner 6020 buckets, inspection of the decompression servo valve, and rectification of oil leakage from the MIV OPU return line filter block inlet flange, along with welding repair of runner buckets. It also involves draining and cleaning of the MIV OPU sump, ultrasonic testing of OPU

accumulators, MIV bypass, and decompression pipelines. In addition, the plan covers flushing of MHE tubes, servicing of the cyclone separator and RCW strainers, as well as overhauling and pre-alignment checks of both CCW Pumps A and B.

3. Turbine System

3.1 Nozzle Injector

Since the nozzle injector system of Unit 4 consists of new GE nozzles installed during the 2022–2023 annual maintenance, all five nozzles were observed to be operating within the acceptable pressure retention range during peak load operation. No oil or water leakage was detected during operation; therefore, no further overhaul is required.

3.2 Runner

1. **TABLE 2 RUNNER PARTICULARS**

Particulars	Runner (6020)
Make	Andritz Hydro, Switzerland
Type	Forged Fabricated
Hard Coating	HVOF
Cumulative Running Hours	13556.44 hours after reclaimed runner installed on unit-4 during 2024-2025AM
Running hours from last inspection	0 Hrs till date (24/02/2026)
Repair/ reclamation History	Dismounted On 11.05.2021 and sent for repair to BHSL on 18.03.2022 after completing 23,080.50 Hrs

The cumulative running hours of Unit IV Runner 6020 since installation is 35,275.26 hours. Erosion, peeling, and pitting were observed in the splitter area of almost all buckets. Major erosion and cracks were identified during runner inspection on bucket numbers 12 (L), 16 (L), 19 (L), and 20 (L). Weld repair was carried out on nearly all the buckets, as most of the bucket surfaces had experienced erosion, followed by Magnetic Particle Testing (MPT) of the runner after the welding work was completed.

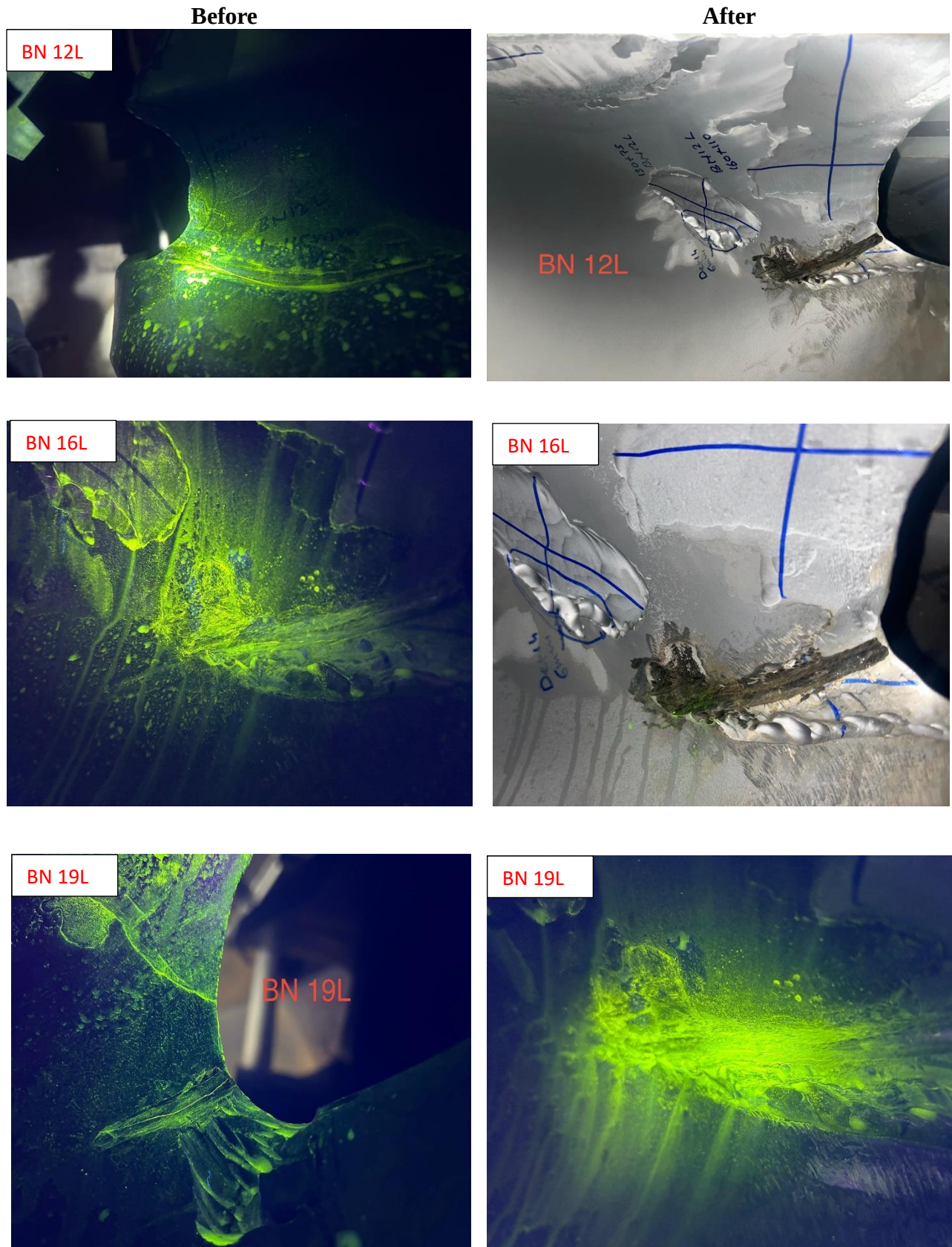


FIGURE 1; RANDOM PICTURES OF

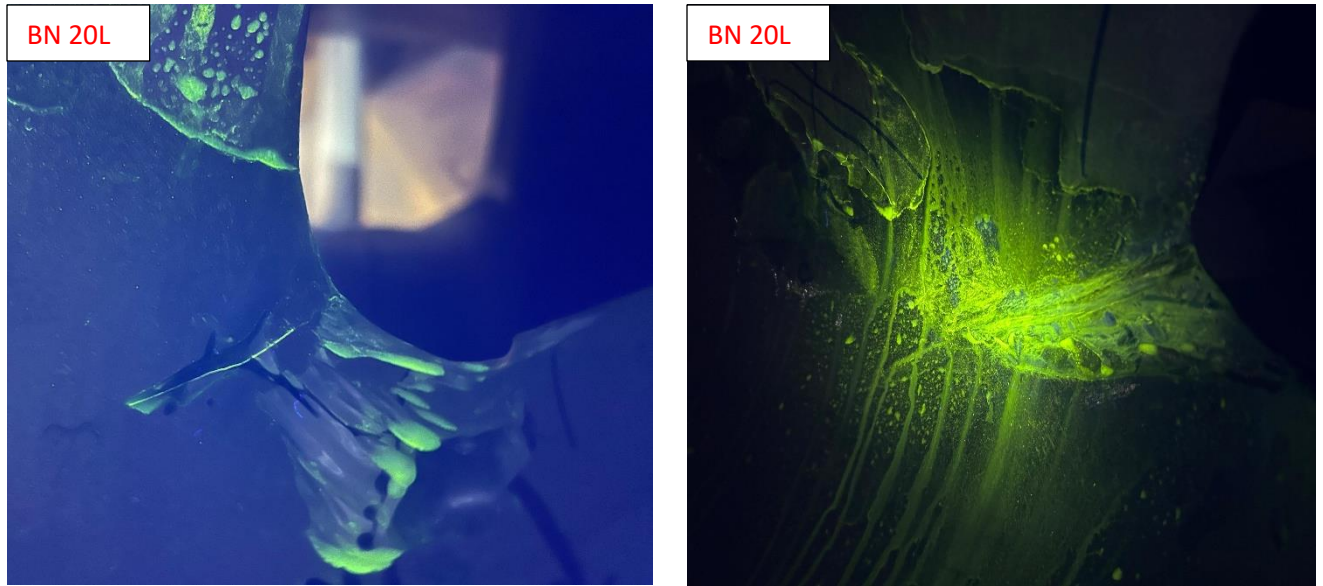


FIGURE 1; BEFORE AND AFTER WELD REPAIR OF RUNNGER BUCKETS





FIGURE 2: Runner bucket grinding on eroded parts

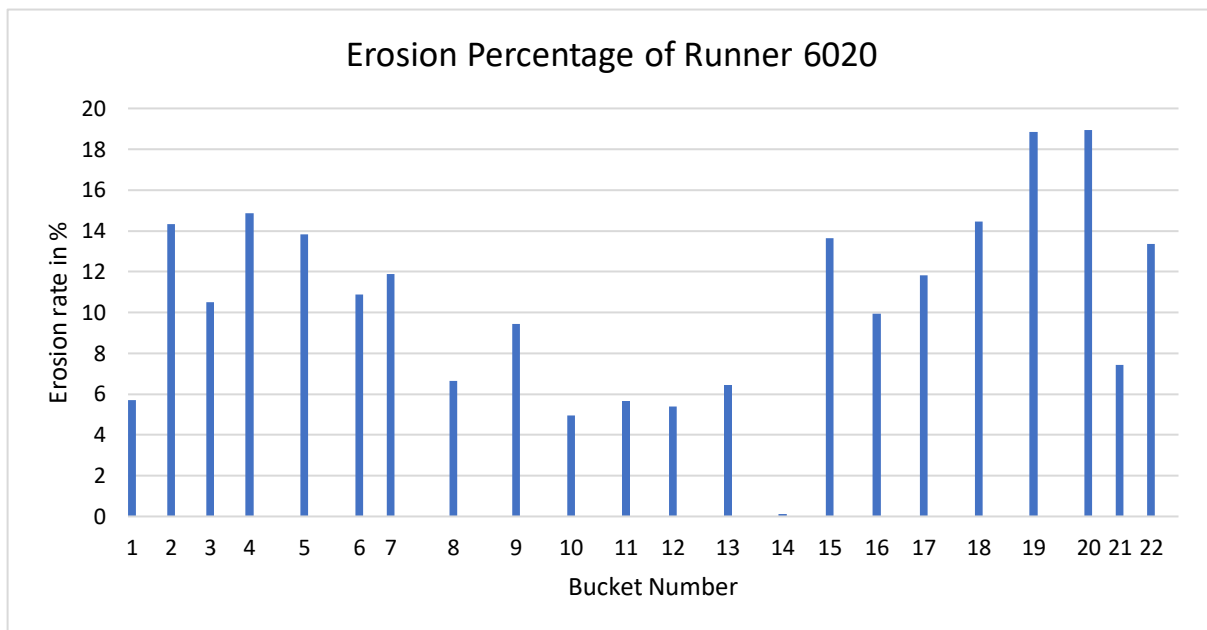


FIGURE 3; EROSION RATE OF EACH BUCKET

The chart illustrates the parent material erosion for each bucket of the runner. The highest erosion is observed on BN 20 with a maximum value of 18.95%, followed closely by BN 19 at 18.83%. The lowest erosion is recorded on BN 14 at only 0.125%. The remaining buckets show erosion values ranging between 6% and 14%, indicating relatively high material erosion across most of the buckets.

4. Governor and MIV system

TABLE 3: SPARES CONSUMED FOR GOVERNOR AND MIV SYSTEM

S. No	Materials	Quantity
1	Filter element-0160 D 020 BN4HC	2
2	Rubber clamps, 80 mm	6
3	S.S flange 100NB for S.S pipe,	2
4	Turbine Oil, VG46(Servo Prime/Turbinol),	130
5	S.S Ball Valve, 1"	1

4.1 Governing System

The main function of governor system is to maintain the rotational speed of turbine within acceptable limits during no-load condition. It performs this function by sensing changes in speed and accordingly controlling flow of water into turbine at fast rate to maintain the speed with reference to grid frequency.

During the pre-inspection and functionality testing of the system, nitrogen gas leakage testing was carried out and the gas pressure was replenished to 64 kg/cm². Pump capacity checking was performed and dry operation of the oil pump was verified locally. Functional checks of the pump, all directional control valves, and visual inspection for leakages were conducted. As the condition and performance of the pumps and idler valve were found satisfactory during operation, no overhaul was carried out as part of the maintenance activities.

As part of the Annual Maintenance 2025–2026, cleaning of the Governor OPU sump and replacement of the pressure line filter between the OPU and VTA were completed. During the pre-check of all GE nozzles, pressure retention was found to be within acceptable limits. Therefore, no nozzle removal or associated control pipeline work was required.



FIGURE 4; GOVERNOR OPU SUMP CLEANING



FIGURE 5; DRAINLINE AND PRESSURE LINE FILTER REPLACEMENT

4.3 MIV System

The turbine provided for shutting off water pressure supply from the pressure shaft/penstock to the turbine distributor during maintenance or emergency closing of the machine. The valve is designed to open under balanced condition and to close against maximum unbalanced head and full turbine pressure of 99 kg/cm² (maximum static head plus head due to water hammer) during emergency. It has a spherical valve with two-body part, door, a downstream service seal and upstream maintenance seal. The valve is operated by a double acting oil operated servomotor which is controlled by a hydraulic operated valve.

During the Annual Maintenance 2025–2026, nitrogen gas leakage testing of the MIV bank was carried out and the gas pressure was replenished. Greasing of the MIV seal valve, penstock drain valve, MIV drain valve, and distributor drain valve was completed. Additionally, decompression servo seating inspection was conducted. Rectification of oil leakage from the MIV return line filter block and flange was carried out.

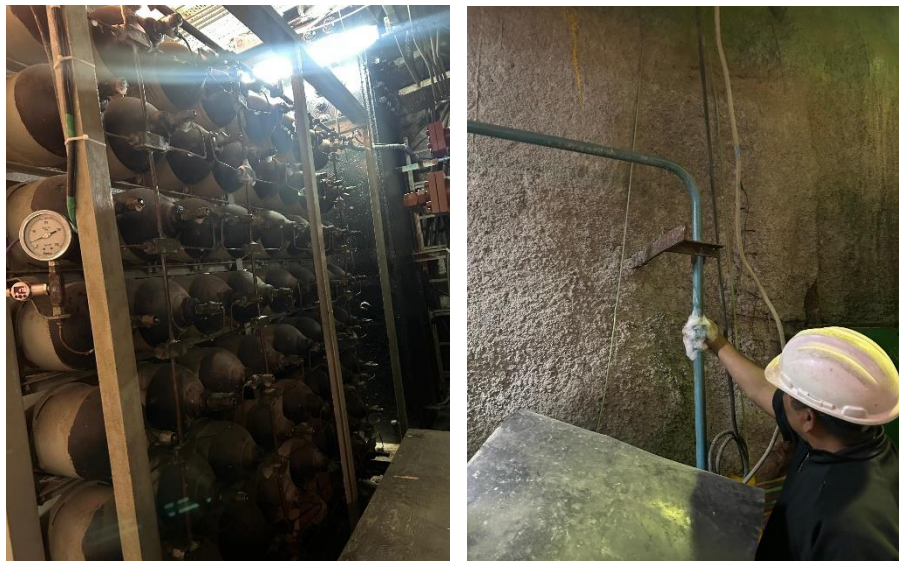


FIGURE 6; NITROGEN BANK AND ITS PIPELINE GAS LEAKAGE TESTING



FIGURE 7; INSPECTION OF DECOMPRESSION SERVO SEATING VALVE FOR ANY FOREIGN DEBRIS

Replacement of the service seal water filter elements was carried out. This activity was mainly undertaken due to the flood that occurred on 10th October 2025, which resulted in a significant accumulation of debris in the system. The filter elements were replaced to prevent potential choking and ensure proper flow in the service seal



FIGURE 8; REPLACEMENT OF SERVICE SEAL FILTER

The calibrated pressure gauges for both the MIV panel and governor control were reinstalled, the N₂ gas for MIV N₂ bank has been replenished up 60Kg and rectification of the MIV OPU cooler isolation valve was also completed. Ultrasonic testing (UT) was carried out on high-pressure pipelines, including the MIV decompression and bypass pipelines. Replacement of service seal water filter elements was also completed as part of the annual maintenance activities.

Rectification of the MIV OPU cooler isolation valve was carried out. Leakage from the MIV heat exchanger end-cap flange was identified and arrested after necessary rectification. The isolation valve at the MIV oil cooler inlet was removed due to leakage, and a crack was observed in the valve housing. Additionally, gasket replacement for the MIV oil cooler outlet isolation valve was carried out, followed by leakage testing to ensure proper sealing.

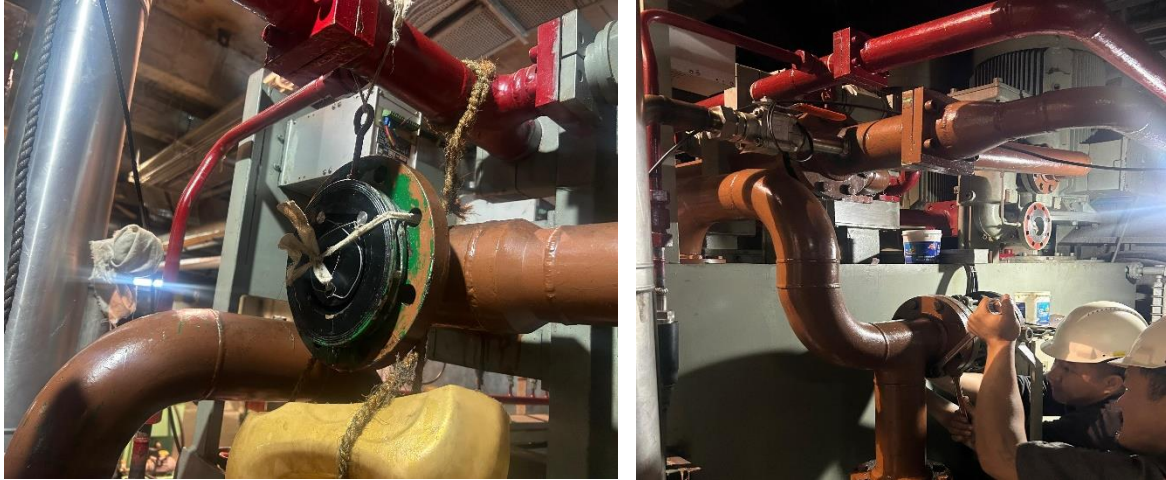


FIGURE 9; RECTIFICATION OF MIV OPU COOLER ISOLATION VALVE AND LEAKAGE ARRESTED

The pressure gauges of the MIV control panel and gauge panel were removed and sent for calibration. After completion of calibration, the gauges were refitted in their respective locations.



FIGURE 10; FITTING BACK THE CALIBRATED PRESSURE GAUGES

Pressure retention testing of the MIV OPU cooler was carried out to check for any tube puncture. During the test, the cooler retained the pressure and no leakage was observed from the tubes. Cleaning and servicing of the MIV OPU were carried out. The oil was replenished after filtration using the LVDH machine as part of the annual maintenance activity.



FIGURE 11; PRESSURE TESTING OF MIV OPU COOLER AND WRAPPING ITS PIPELINE

The ultrasonic testing of Decompression pipeline, bypass pipeline and pressure accumulators checked for any possible deviations in thickness and cracks by the NDT personnel from HRDC to ensure safe parameters of the control pipelines for safe and smooth operations.



FIGURE 12; ULTRASONIC TESTING(UT) ON HIGH Pr. PIPELINES AND ACCUMULATORS

5. Cooling Water System

5.1 Clean Cooling Water System

Two circulating cooling water (CCW) pumps, one operating and one standby, are provided to supply clean cooling water to the generator stator, bearing coolers, and lubrication system. During the inspection, both pumps were checked for proper operation and were found to be in good and healthy condition.

Pre-alignment checks were carried out to verify that there was no disturbance in the base foot positioning and to ensure proper alignment between each pump and its motor. Since both pumps were operating satisfactorily and their condition was found to be healthy, overhaul servicing was not considered necessary.

The tightness and overall condition of the complete system were also inspected and found to be normal. All electrical terminations were checked for looseness and tightened wherever required.



FIGURE 13; CCW pump alignment check

5.2 Raw Water-Cooling System

Two vertical turbine pumps, one designated as the main pump and the other as the standby pump, are installed to supply cooling water to the heat exchangers, generator transformer, and shunt reactor.



FIGURE 14: Cleaning and servicing of RCW strainers

During the maintenance shutdown, the motorized water strainer was completely overhauled, thoroughly cleaned, and reassembled. A significant amount of debris was found accumulated on the strainer cones. It is recommended that the strainer cones be cleaned during every annual maintenance outage to prevent excessive debris buildup and blockage of the outlet holes. Additionally, the non-return valves (NRVs) of both RCW pumps were inspected and found to be in satisfactory condition.

5.3 Cyclone Separator

The cyclone separator, which is used to remove coarse sediments from the raw water supplied for transformer cooling, was overhauled, cleaned, and serviced during the maintenance period. Sediments accumulated at the bottom during operation were removed manually and by flushing.



FIGURE 15: Flushing of sediments from cyclone separator

5.4 Main Heat Exchanger (MHE)

A Main Heat Exchanger (MHE) is installed in each unit to cool the clean circulating water used in the generating system for the stator air coolers, OPU coolers, and bearing coolers (UGB, LGB, and TGB). Raw water flows through the heat exchanger tubes to ensure efficient heat transfer.

During the maintenance shutdown, the heat exchanger tubes were thoroughly cleaned and flushed. A hydrostatic pressure test was subsequently carried out at 7 kg/cm² for 30 minutes, and no leakage was observed. The test result is attached as annexure.



FIGURE 16: Removal of old MHE and mounting with new MHE

Annexure

S.NO	Annexure	Description
1	Annexure 1	Bypass Timing set
2	Annexure 2	Underwater component inspection
3	Annexure 3	Unit-4 Runner MPT report
4	Annexure 4	Pre-checklist Turbine
5	Annexure 5	Governor checklist cum OK card
6	Annexure 6	Final checklist Turbine
7	Annexure 7	Final checkpoints before spinning
8	Annexure 8	Pre-checklist Cooling water system
9	Annexure 9	Final Checklist of Cooling water system
10	Annexure 10	Final checklist governor
11	Annexure 11	Check points before carrying out repair work in turbine pit
12	Annexure 12	Deflector gap setting
13	Annexure 13	Deflector servo timing set
14	Annexure 14	Deflector-nozzle relation
15	Annexure 15	Gov N2 pressure retention check
16	Annexure 16	Gov OPU pump capacity check
17	Annexure 17	Gov OPU safety valve setting
18	Annexure 18	Gov system functionality check
19	Annexure 19	Gov system pressure retention check
20	Annexure 20	MIV checklist cum OK card
21	Annexure 21	MIV OPU safety valve setting
22	Annexure 22	MIV pressure retention check
23	Annexure 23	Nozzle opening closing timing check
24	Annexure 24	MIV system timing set
25	Annexure 25	MHE pressure testing and CCW pump alignment
26	Annexure 26	

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